Block practice · Series and the binomial expansion

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Least or greatest number of terms for an arithmetic sum

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Practice workbook · Least or greatest number of terms for an arithmetic sum

Setting an arithmetic sum greater or less than a target, solving the quadratic inequality and choosing the right whole number.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-07

1. Question 2 · part whole · 7 marks

2025 JUN · Paper 01 (regional R)

Worked-example retry · Attempt part whole only.

2 An arithmetic series F has third term 8 and fifth term 20

Given that the sum to n terms of F is greater than 220

find the least value of n

(7)

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1. Question 2 · part whole · 7 marks

2025 JUN · Paper 01 (regional R)

Worked-example retry · Attempt part whole only.

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Question 2 continued

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(Total for Question 2 is 7 marks)

1. Question 2 · part whole · 7 marks · Working and answer

Working and answer

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2. Question 6 · part b · 3 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

6 (a) Show that $\sum_{r=1}^{n}(4r-3) = n(2n-1)$

(3)

(b) Hence, or otherwise, find the least value of n such that $\sum_{r=1}^{n}(4r-3) > 1000$

(3)

Given that $S_n = n(2n-1)$, $t_n = (4n-3)$ and that $18 + 3t_{n+7} = S_{n+4}$

(c) find the value of n.

(4)

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2. Question 6 · part b · 3 marks · Working and answer

Working and answer

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3. Question 4 · part b.ii · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part b.ii only.

4 Given that $\sum_{r=1}^{n}(4r+A)=\frac{n}{2}(4n-6)$ where A is a constant

(a) show that $A=-5$

(3)

Hence, or otherwise,

(b) (i) evaluate $\sum_{21}^{60}(4r-5)$

(2)

(ii) find the greatest value of n such that

$$\sum_{r=1}^{n}(4r-5)<3418$$

(3)

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3. Question 4 · part b.ii · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part b.ii only.

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Question 4 continued

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3. Question 4 · part b.ii · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part b.ii only.

Question 4 continued

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3. Question 4 · part b.ii · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part b.ii only.

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Question 4 continued

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(Total for Question 4 is 8 marks)

3. Question 4 · part b.ii · 3 marks · Working and answer

Working and answer

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4. Question 3 · part whole · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

3 An arithmetic series has first term 16 and common difference –5

The sum to n terms of this series is S_n

Given that $S_n < -450$

find the least value of n

(4)

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4. Question 3 · part whole · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 4 marks)

4. Question 3 · part whole · 4 marks · Working and answer

Working and answer

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